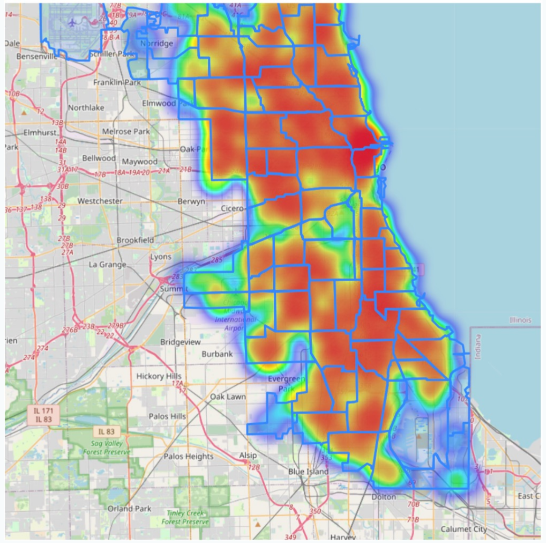
Beyond Crime Counts

Chicago Neighborhood Safety Profiles, 2020-2024

Target audience :
Chicago public safety planners, community development analysts, and neighborhood organizations



Decision use :
Move from raw incident counts to a profile-based view: volume + severity mix + socioeconomic context

Main message: total reported crime count alone is not enough to describe neighborhood safety.

Python provides the main EDA story; SQL validates, drills down, and extends it to additional socioeconomic indicators.

Study design: neighborhood risk as a multidimensional profile

The target audience needs actionable neighborhood categories, not just a list of highest-count areas.

Data sources

1. City of Chicago crime incidents, 2020-2024 (government data)
2. Socioeconomic indicators by community area (government data)

Workflow

Python: cleaning, EDA, plots, modeling

SQL: validation, joins, filters, rankings, grouped comparisons

Outcome Measures

- Total reported crime volume
- Violent-crime share
- Domestic-related share
- Community risk profile

How the results help the audience

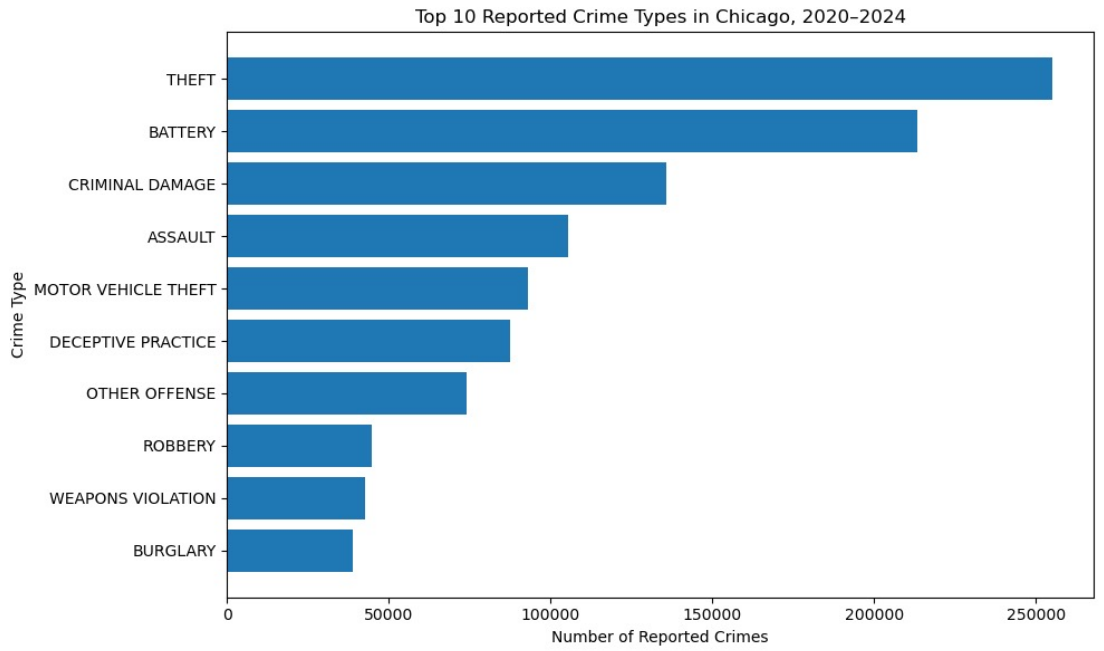
Separate high-activity areas from high-severity areas.

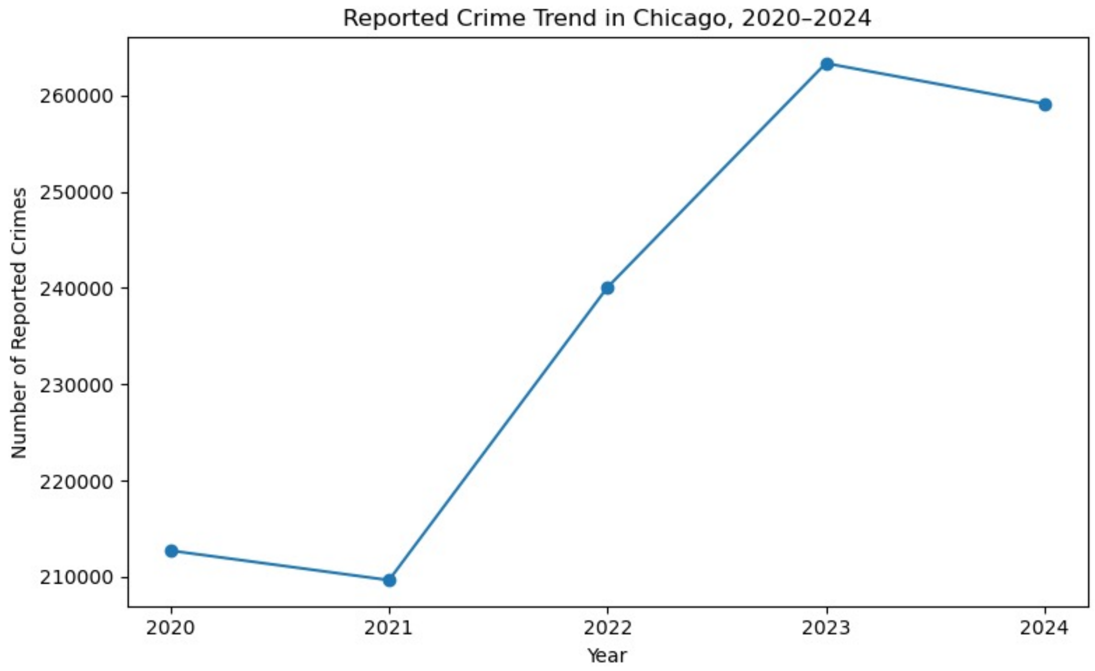
Identify communities that could be missed by total-crime rankings.

Connect crime composition to socioeconomic context for more tailored interventions.

Overview: citywide patterns set the context

Theft and battery dominate reported incidents; total volume rose after 2021 and remained elevated in 2024.

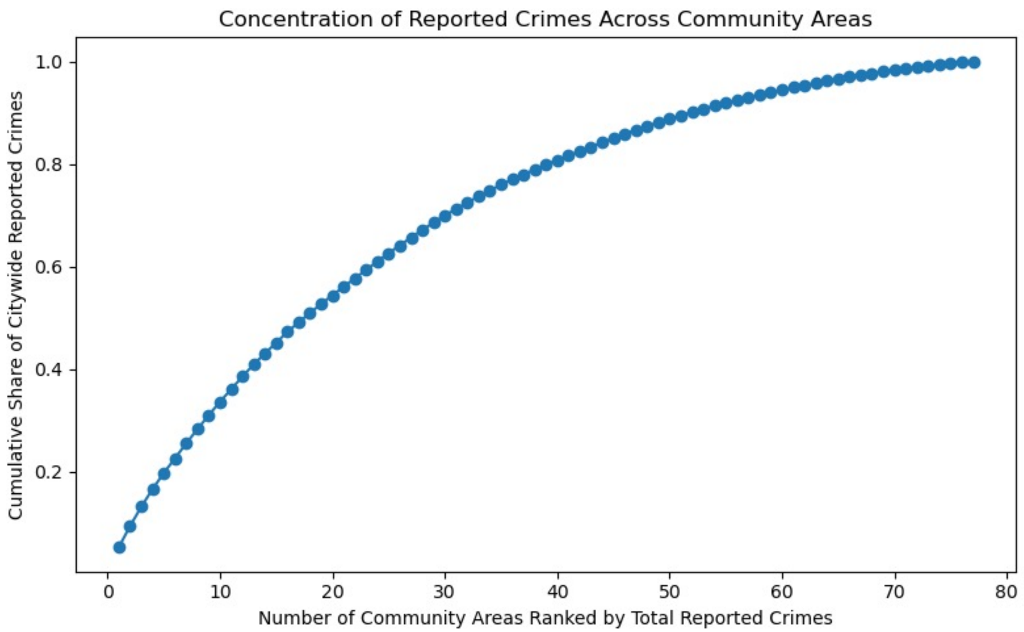




Citywide context: reported crime is shaped by both crime-type composition and post-2021 growth.

Finding 1: reported crime is concentrated, but high-volume areas differ

Python shows concentration visually; SQL explains why high-volume areas are not one single type.



SQL Q4: top 10 high-crime communities split by hardship type

Group	N	Avg crimes	Violent %	Domestic %	Hardship	Poverty %	Unemp. %	Income
High-volume / higher-hardship	6	39,004	37.0	26.6	73.3	32.3	22.1	$15,664
High-volume / lower-hardship	4	41,153	24.5	8.0	7.3	15.7	7.5	$60,521

SQL Q5: all above-average crime communities show the same split

Group	N	Avg crimes	Violent %	Domestic %	Arrest %	Hardship	Poverty %	Unemp. %	No HS %	Income
High-crime / higher-hardship	20	26,984	36.6	25.8	15.8	75.0	31.4	22.2	25.6	$15,253
High-crime / lower-hardship	10	28,976	25.1	10.0	10.9	16.4	16.8	7.4	10.0	$48,837

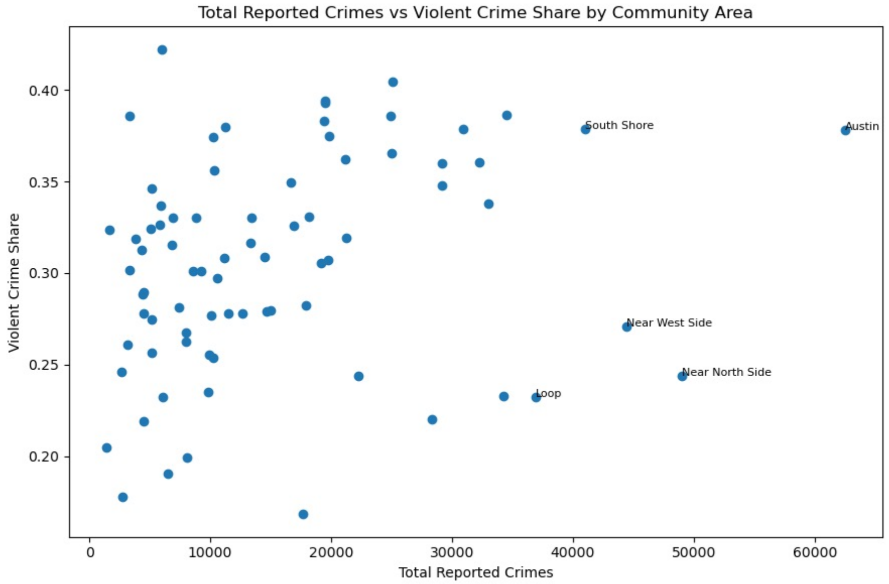
Python EDA: the cumulative curve rises quickly, meaning a relatively small group of community areas accounts for a large share of reported incidents.

SQL drill-down: among high-crime communities, similar volume can split into higher-hardship vs lower-hardship profiles. Higher-hardship high-crime areas show much higher violent and domestic shares.

Audience use: avoid treating all high-count areas as the same type of public safety problem

Finding 2: crime volume and violent share measure different risks

Python quantifies the weak relationship; SQL identifies the neighborhoods behind the split.



SQL Q7: high volume, lower violent share

Community	Crimes	Violent share	Hardship	Poverty %	Income
Near North Side	49,009	0.244	1.0	12.9	$88,669
Near West Side	44,421	0.271	15.0	20.6	$44,689
Loop	36,932	0.232	3.0	14.7	$65,526
West Town	34,249	0.233	10.0	14.7	$43,198
Lake View	28,361	0.220	5.0	11.4	$60,058
Logan Square	22,217	0.244	23.0	16.8	$31,908
Uptown	19,193	0.305	20.0	24.0	$35,787
West Ridge	17,912	0.282	46.0	17.2	$23,040
Lincoln Park	17,696	0.169	2.0	12.3	$71,551

SQL Q8: lower volume, higher violent share

Community	Crimes	Violent share	Hardship	Poverty %	Income
Riverdale	5,948	0.422	98.0	56.5	$8,201
Fuller Park	3,322	0.386	97.0	51.2	$10,432
Washington Park	11,239	0.380	88.0	42.1	$13,785
Brighton Park	10,278	0.374	84.0	23.6	$13,089
Gage Park	10,291	0.356	93.0	23.4	$12,171
Armour Square	5,185	0.346	82.0	40.1	$16,148
East Side	5,919	0.337	64.0	19.2	$17,104
South Deering	8,852	0.330	65.0	29.2	$14,685
Hermosa	6,885	0.330	71.0	20.5	$15,089
Lower West Side	13,397	0.330	76.0	25.8	$16,444

SQL Q7 names high-activity areas with lower severity mix: Near North Side, Loop, West Town, Lake View, and Lincoln Park.

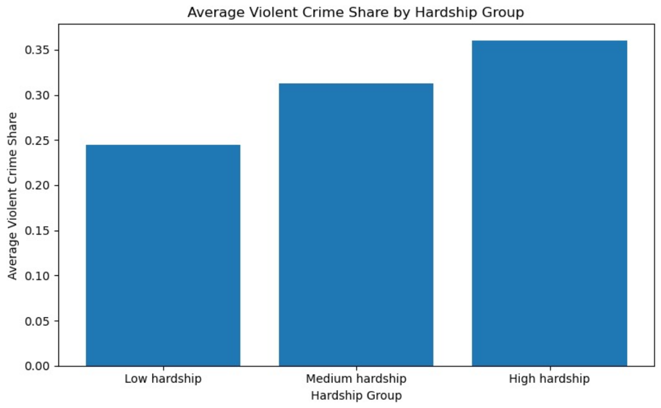
SQL Q8 names hidden-severity communities: Riverdale, Fuller Park, Washington Park, Brighton Park, and Gage Park.

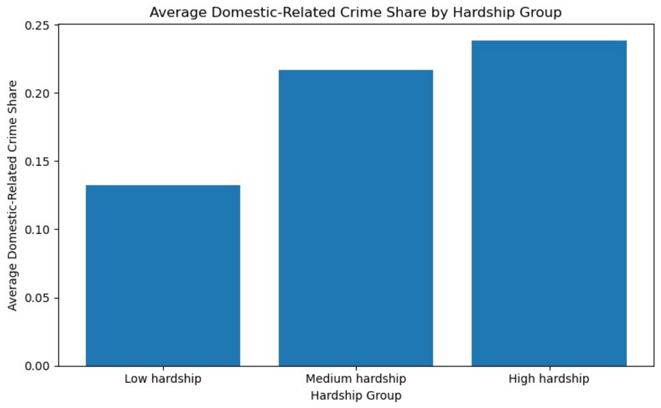
Python EDA: total crime count and violent-crime share do not move together strongly. This means ranking by incident count alone can misread neighborhood risk.

Audience use: use volume + severity together to avoid missing hidden high-severity communities

Finding 3: hardship aligns more clearly with crime composition than raw volume

Python charts and SQL grouped aggregation both show the same gradient.





Python: violent-crime share and domestic-related share rise across hardship groups.

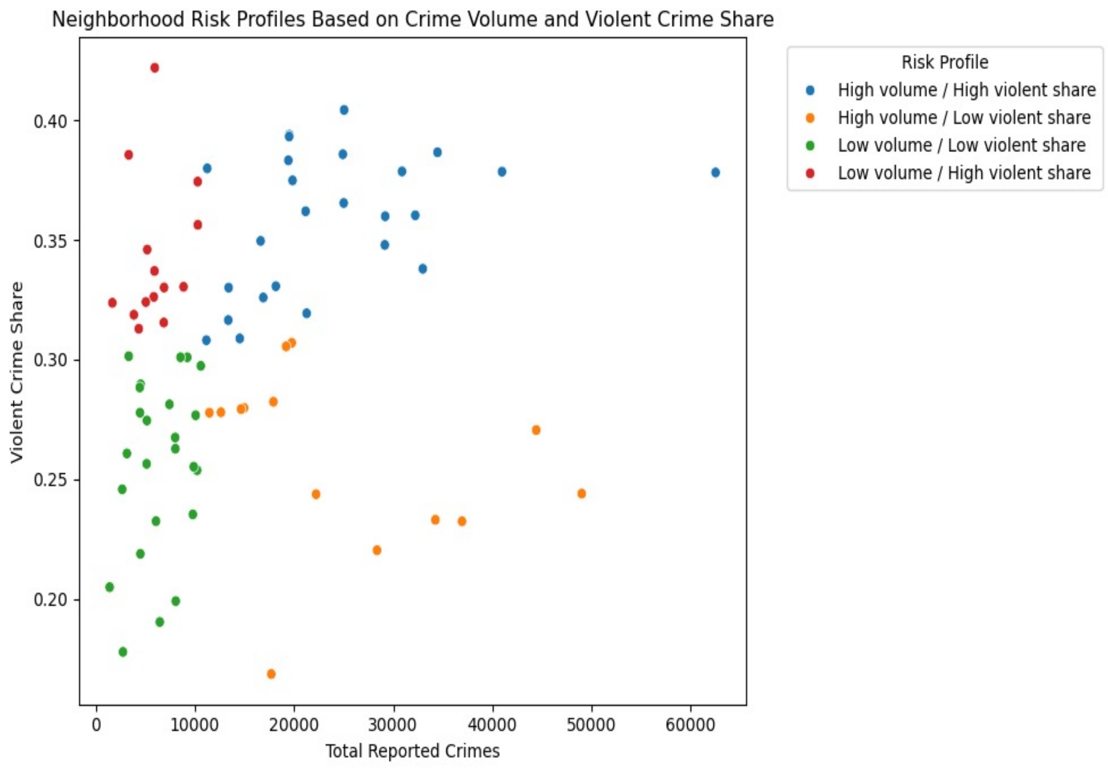
SQL Q6: the same pattern appears after joining crime summaries with broader socioeconomic indicators. High hardship also corresponds to higher poverty, unemployment, no-HS-diploma rate, crowded housing, and lower income.

Group	N	Avg crimes	Violent %	Domestic %	Arrest %	Hardship	Poverty %	Unemp. %	No HS %	Income
Low Hardship	38	13,625	26.1	15.2	10.1	24.5	13.8	10.2	12.1	$35,914
Mid Hardship	19	17,259	32.7	23.2	12.0	61.1	22.6	17.7	25.3	$17,905
High Hardship	20	16,961	36.9	24.4	15.1	86.0	36.1	23.0	31.3	$13,173

Audience use: combine socioeconomic context with crime composition when prioritizing community support

Finding 4: risk profiles translate EDA into actionable community types

Python builds the typology; SQL filters validate the same split.



Profile framework for target audience

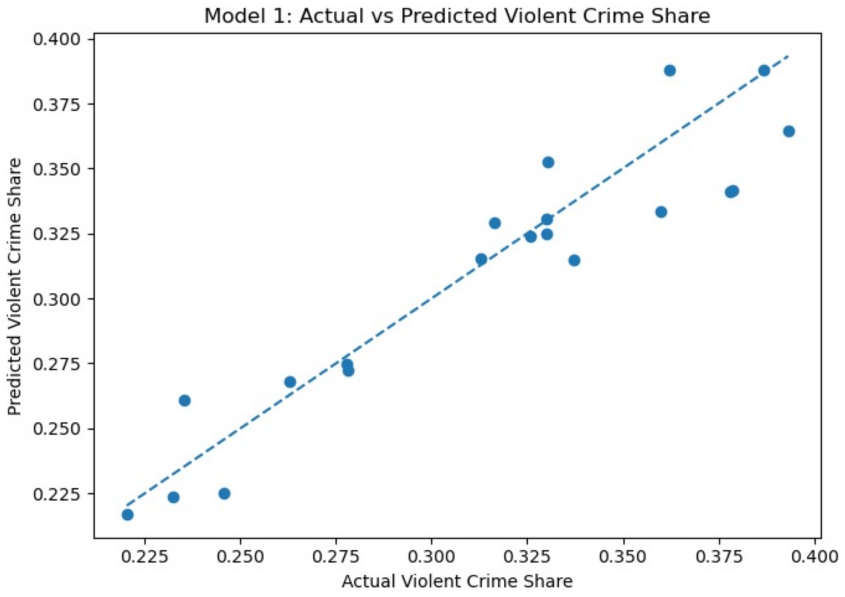
High volume / high violent share	Austin; North Lawndale; Englewood	Highest combined risk
High volume / low violent share	Loop; Near North Side; Near West Side	High activity, lower severity mix
Low volume / high violent share	Riverdale; Fuller Park	Hidden risk if ranked only by count
Low volume / low violent share	Forest Glen; Edison Park; North Center	Lower-risk profile

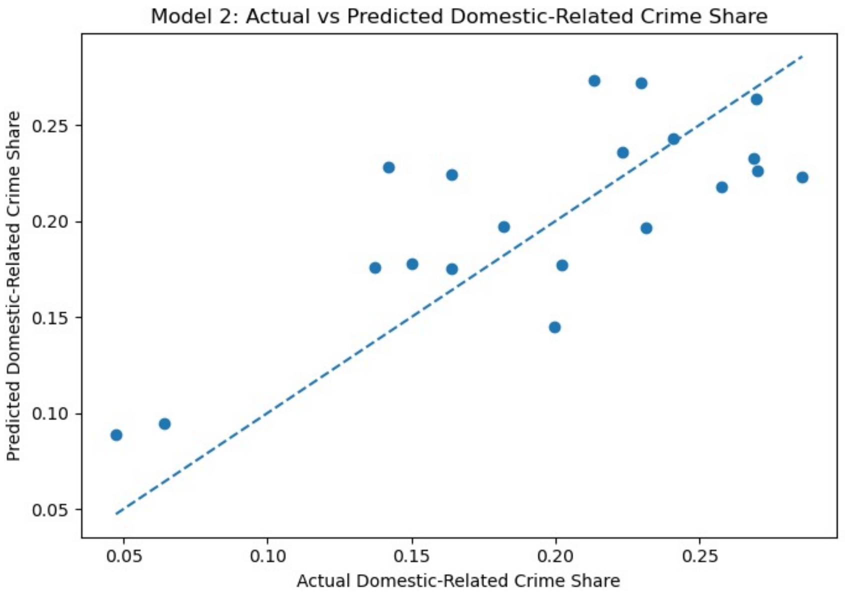
SQL match: Query 7 isolates high-volume / lower-violent-share communities; Query 8 isolates low-volume / higher-violent-share communities. This confirms the Python profile types using relational filtering, not just visualization.

Audience use: match intervention type to profile type instead of using one citywide rule

Modeling check: socioeconomic indicators explain meaningful variation

Models support the EDA patterns; the interpretation remains descriptive, not causal.





Violent-crime-share model
$R^2 \approx 0.87$
Socioeconomic indicators strongly align with violent-crime share across community areas.

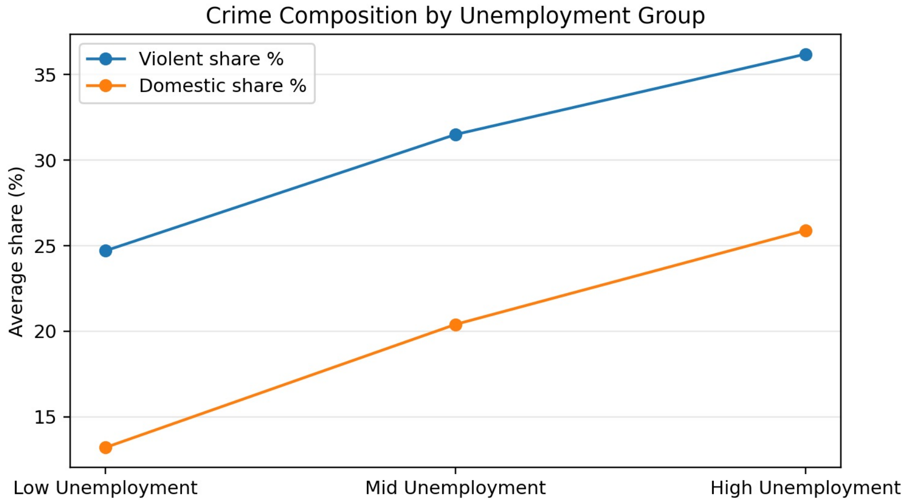
Domestic-related-share model
$R^2 \approx 0.58$
The relationship is weaker but still meaningful for descriptive profiling.

Audience use: the model is not used to claim causality. It supports a screening framework: socioeconomic context can help identify where crime composition is more severe.

Additional SQL Investigation: Higher Unemployment Aligns with Higher Violent/Domestic Shares

SQL Q11 extends the Python analysis by testing unemployment as an additional socioeconomic indicator.

Group	N	Avg crimes	Violent %	Domestic %	Arrest %	Unemp. %	Poverty %	Income	Hardship
Low Unemployment	25	14,177	24.7	13.2	10.0	7.7	13.0	$39,955	20.3
Mid Unemployment	31	12,591	31.5	20.4	11.8	14.8	21.7	$19,938	58.6
High Unemployment	21	20,959	36.2	25.9	14.2	25.3	32.2	$16,733	70.8



SQL Q11 finding: as unemployment group rises, violent-crime share increases from 24.7% to 31.5% to 36.2%; domestic-related share increases from 13.2% to 20.4% to 25.9%.

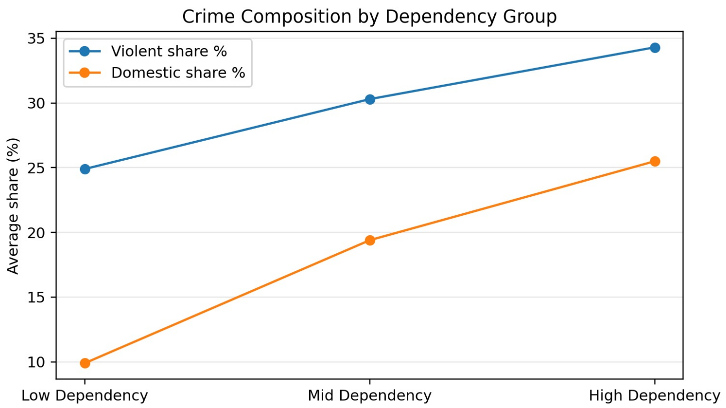
Interpretation: unemployment is not treated as a causal factor. It is used as an additional socioeconomic signal that helps flag communities where the crime mix is more severe.

Audience use: use unemployment as an additional screening indicator when identifying communities with higher violent/domestic shares

Additional SQL Investigation: Higher Dependency Also Aligns with More Severe Crime Mix

SQL Q12 tests age dependency separately; SQL Q13 combines unemployment and dependency to identify overlapping vulnerability.

Group	N	Avg crimes	Violent %	Domestic %	Arrest %	Dep. %	Unemp. %	Poverty %	Income	Hardship
Low Dependency	14	23,179	24.9	9.9	10.3	22.7	7.3	15.7	$49,392	12.9
Mid Dependency	39	12,253	30.3	19.4	11.6	36.4	14.4	19.1	$21,495	52.1
High Dependency	24	15,939	34.3	25.5	13.2	42.4	21.6	29.6	$18,273	66.7



Group	N	Avg crimes	Violent %	Domestic %	Unemp. %	Dep. %	Poverty %	Income	Hardship
High unemployment / High dependency	31	17,727	35.3	24.5	22.8	41.0	31.0	$15,888	73.3
High unemployment / Lower dependency	5	19,153	34.4	21.0	17.1	33.6	27.8	$21,189	60.0
Lower unemployment / High dependency	17	7,760	27.8	20.0	10.9	39.1	13.1	$25,000	41.5
Lower unemployment / Lower dependency	24	16,987	25.6	12.6	8.6	27.0	14.8	$39,371	22.3

SQL Q12 finding: as dependency group rises, violent-crime share increases from 24.9% to 30.3% to 34.3%; domestic-related share increases from 9.9% to 19.4% to 25.5%.

SQL Q13 finding: communities with both high unemployment and high dependency have the highest violent-crime share (35.3%) and domestic-related share (24.5%), suggesting overlapping socioeconomic and demographic vulnerability.

Audience use: combine labor-market stress and demographic vulnerability to refine community risk profiles

Conclusion: neighborhood safety is multidimensional

For public safety planners, the findings turn EDA into a practical screening framework.

Main findings

- Crime volume alone is incomplete
- Violent-crime share captures severity
- Socioeconomic context helps explain risk profiles

Python:
- Found concentration and composition patterns
- Built the risk-profile framework

SQL:
- Validated 1.18M+ records and 77 areas
- Named communities behind each pattern
- Extended analysis to unemployment and dependency

1. For Audience: Screen by volume

Identify where reported incidents concentrate.

2. For Audience: Add severity

Use violent-crime share to avoid missing hidden high-severity communities.

3.For Audience: Add socioeconomic context

Use hardship, unemployment, dependency, education, housing, and income to refine intervention type.

Limitations and future work

- Descriptive associations, not causal effects
- Counts are not population-normalized
- Future work: crime rates per 1,000 residents, land use, transit, business activity, police staffing, and mobility data